

Chapter-7 Data Acquisition System

1. What is data acquisition system?

Ans: Data acquisition system is a device or an integrated system which collects information of the state or condition of various parameters of the process.

2. Explain in brief about the analog data acquisition technique (system).

Ans:

→ Analog data acquisition system is the data acquisition system in which data is obtained in the analog form.

→ The different components of analog data acquisition system is shown in the figure.

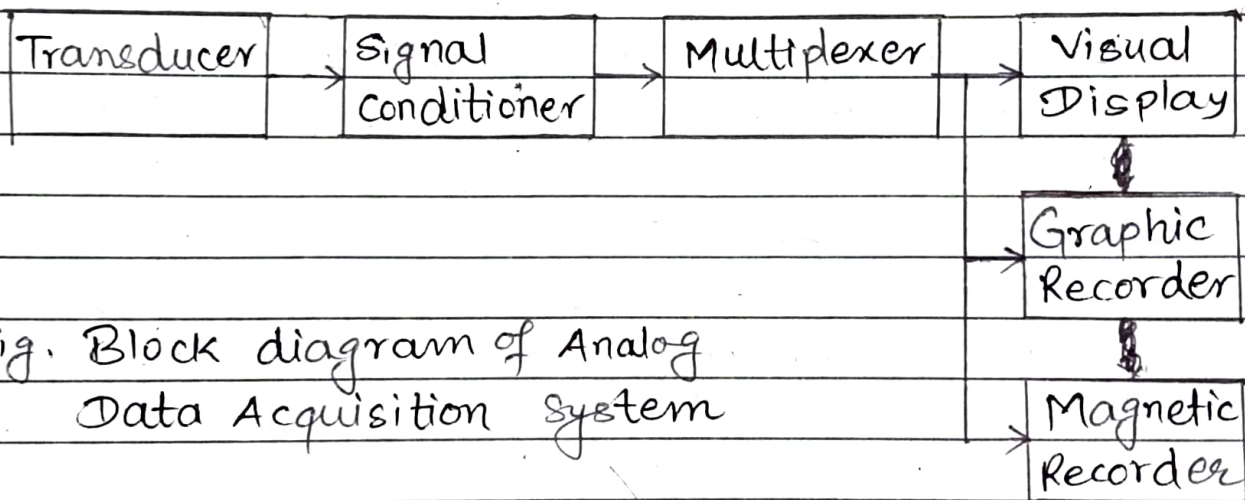


Fig. Block diagram of Analog Data Acquisition system

i) Transducer:

→ Transducer is a device used to convert physical parameters (i.e. pressure, temperature) into electrical signals.

ii> Signal Conditioner:
→ Signal Conditioner is an equipment used for amplifying, refining or selecting certain portions of electrical signals.

iii> Multiplexer:
→ Multiplexer is a device used for scanning different signal sources.

iv> Visual Display:
→ Visual display is a device used for continuous monitoring of the signal.

v> Graphic Recorder:
→ Graphic recorder is a device used for obtaining permanent records of input data.

vi> Magnetic Recorder:
→ Magnetic recorder is a device used for preserving and reproducing data.

4. Briefly explain about the digital data acquisition system.

Ans: A digital data acquisition system is a system that converts analog signal into digital signals for processing, storage and analysis.

→ The digital data acquisition system consists of different components as shown in figure:

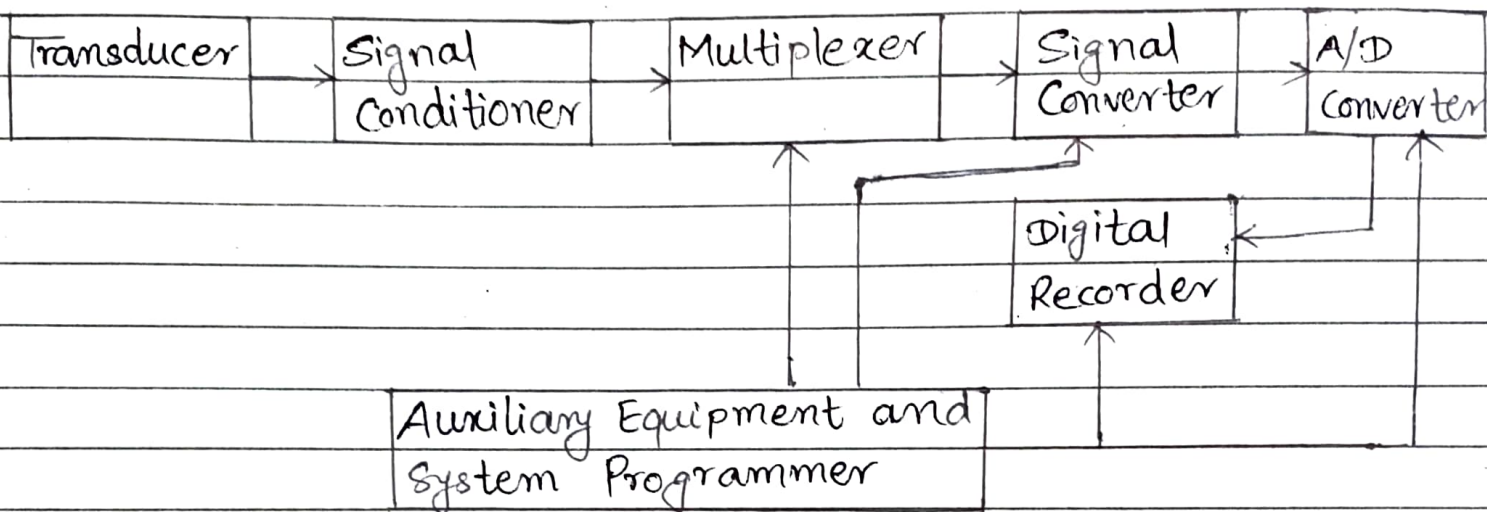


Fig. Digital Data Acquisition System

A digital data acquisition system essential functional operations includes:

- Handling of analog signal.
- Carrying out of measurement of analog signals.
- Necessary conversion and handling of digital data.
- Internal Programming and control

The different Components of ~~data~~ digital data acquisition system are as discussed below:

i> Transducer:

→ Transducer is a device used to convert physical parameters (temperature, pressure) into electrical signal.

ii> Signal Conditioner:

→ Signal Conditioner is an equipment used for amplifying, refining or selecting certain portion of electrical signals.

iii> Multiplexer:

→ Multiplexer is a device used for scanning different signal resources.

iv> Signal Converter:

→ Signal converter converts analog signal to suitable form that becomes acceptable by A/D Converter.

v> A/D Converter:

→ A/D Converter converts analog voltage into its equivalent digital form.

vi> Digital Recorder:

→ Digital recorder is used to record digital information on punch cards, perforated paper tape, magnetic tape etc.

vii> Auxiliary Element:

→ Auxiliary element performs some typical functions like linearization and limit comparison of signals and these functions get performed by individual instruments or by a digital computer.

5. What are the uses of data acquisition system?

Ans Data acquisition system is a device or an integrated system which collects the information of the state or condition of the various parameters of the process.

Some uses of data acquisition systems are as follows:-

i> Scientific Research:

→ In Scientific research, data acquisition systems are used to collect experimental data.

ii> Industrial Monitoring:

→ In industrial monitoring Data Acquisition systems help to monitor and control processes in manufacturing.

iii> Medical Diagnostics:

→ Data acquisition system helps to collect data from medical devices such as EEGs, ECGs and MRIs.

iv> ~~Scie~~ Environ

iv> Environmental Monitoring:

→ Data acquisition system helps to collect data from environment sensors such as air quality monitors and weather stations.

v> Military applications:

→ Data acquisition system helps to collect data from sensors such as radar and sonar.

6. Differentiate between analog DAS and digital DAS.

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S.N.	Analog DAS	S.N.	Digital DAS
i)	Analog DAS stands for Analog Data Acquisition System.	i)	Digital DAS stands for Digital Data Acquisition System.
ii)	The basic components used in analog DAS are: 1. Transducer 2. Signal Conditioner 3. Multiplexer 4. Visual Display 5. Graphic Recorder 6. Magnetic Recorder	ii)	The basic components used in Digital DAS are: 1. Transducer 2. Signal Conditioner 3. Multiplexer 4. Signal Converter 5. A/D converter 6. Digital Recorder 7. Auxiliary Equipment
iii)	Analog DAS does not need signal conversion.	iii)	Digital DAS needs signal conversion.
iv)	Analog DAS is used when wide bandwidth is required.	iv)	Digital DAS is used when narrow bandwidth is required.
v)	Analog DAS has relatively less accuracy than digital DAS.	v)	Digital DAS has relatively high accuracy than analog DAS.
vi)	Analog DAS has high cost per channel.	vi)	Digital DAS has low cost per channel.

Modern trend in data acquisition system;

→ The generalized block diagram of modern trend in data acquisition system is shown below;

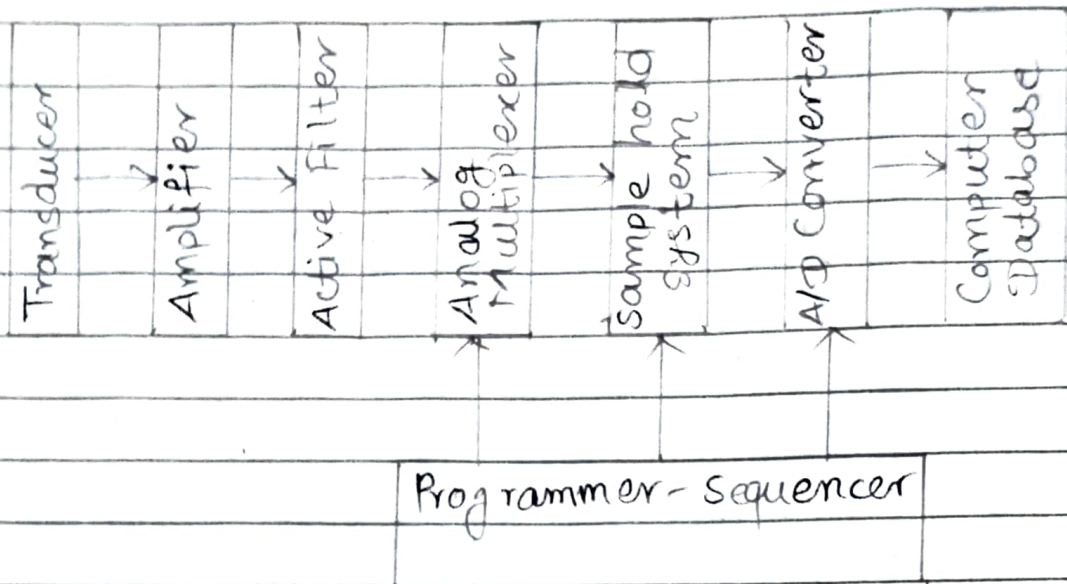


Fig. Block diagram of modern trend in data acquisition system

i) Transducer:

→ Transducer is a device used to convert physical parameters like pressure, temperature into electrical signals.

ii) Amplifier:

→ Amplifier is a device which converts low input signal into suitable output signal for further processing.

iii) Active Filter:

→ The amplifier is frequently followed by low pass filter which reduces the high frequency signal components, i.e. non-electric or electronic noise from the signal.

iv) Analog Multiplexer:

→ The processed analog signal next goes to the analog multiplexer which selects and switches between different analog input channels.

v) Sample hold system, A/D Converter and Computer Database:

→ The Analog multiplexer, together with the sample hold system shares the A/D converter with a number of analog input channels.

→ The timing and control of complete data acquisition system is done by a digital circuit called programmer sequence which is under the control of the system.

→ In some cases, computer itself may control the entire data acquisition system.

1. Define output device with example.

Ans → The data presentation devices are called output devices.
 → The output devices are not only display devices for indicating the output but also used as control devices examples: speaker, monitor, printer, projector, plotter etc.

2. Explain the principle operation of x-y recorder with its application area.

Ans:

- X-Y recorder is an instrument which gives a graphic record of the relationship between two variables.
- There are two types of X-Y recorder: Analog X-Y recorder and digital X-Y recorder.
- The X-Y recorder plots one voltage as a function of another voltage.

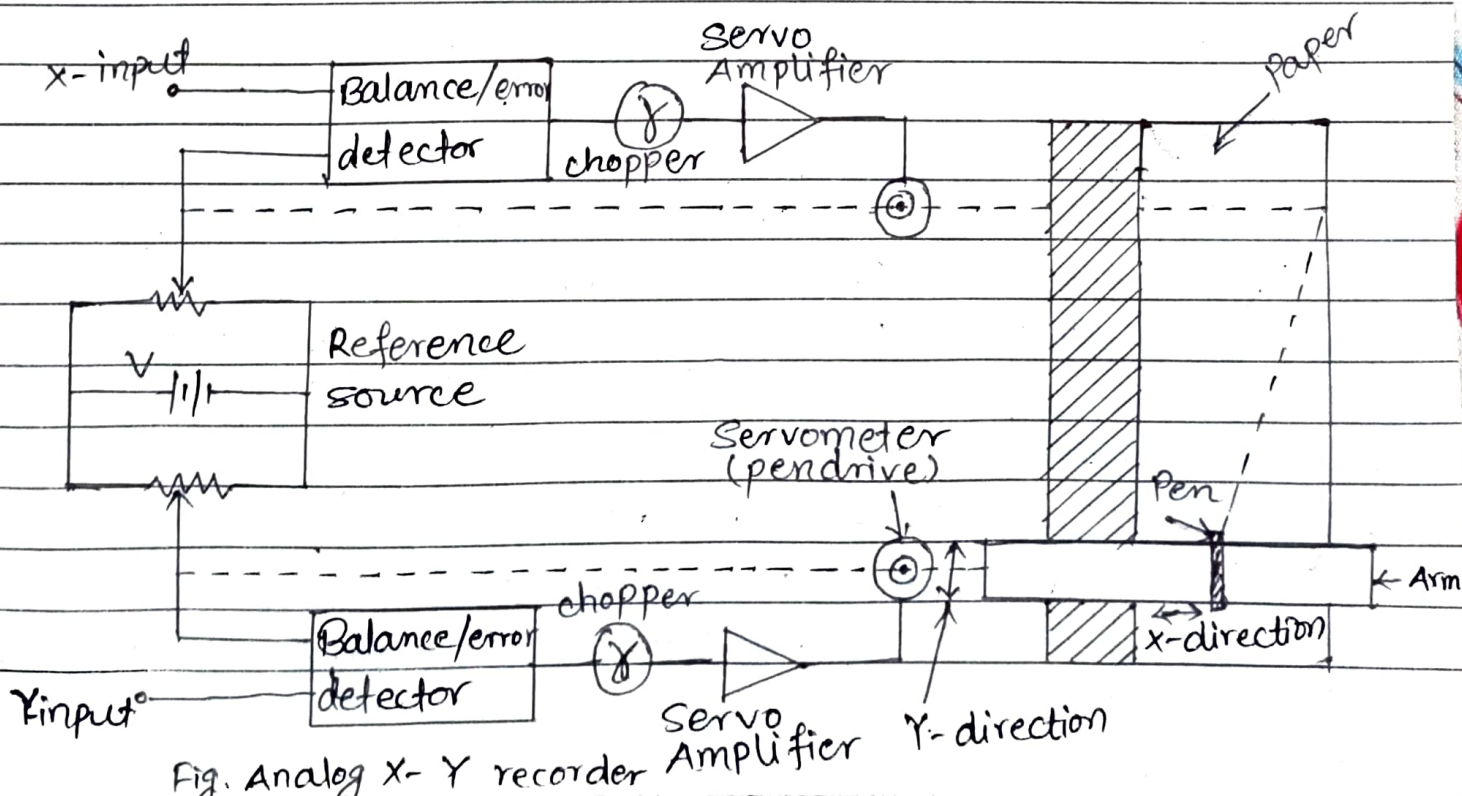


Fig. Analog X-Y recorder

- A signal enters in each of the two channels.
- The signals are attenuated to the inherent full scale range of the recorder.
- The signal then passes to a balance circuit where it is compared with an internal reference voltage.
- The balance circuit compares attenuated signal to a fixed reference voltage.
- The output of the error detector is a difference between the variation in input signal and reference voltage.
- The balancing circuit and error detector gives error signal i.e. DC signal.
- The chopper converts error signal into AC signal.
- The AC signal is then applied to the servo amplifier.
- The servo amplifier drives servomotor which drives writing assembly on a fixed graph paper.
- There are two circuits for two different inputs to be recorded.
- The same action takes place in both axes simultaneously.

* Application areas of X-Y recorder:

- i> Plotting of stress-strain curves, hysteresis curves.
- ii> Plotting speed-torque characteristics of motor.
- iii> Plotting pressure flow studies for lungs.
- iv> Regulation curves of materials.
- v> Pressure-volume diagrams of IC engines.
- vi> Plotting characteristics of vacuum tubes, zener diodes, rectifiers, transistors etc.

3. Explain the magnetic tape recorder with its application.

Ans

→ The magnetic tape recorders are used for high frequency signal processing.

→ The basic components of magnetic tape recorders are:

a) Recording Head:

→ Recording head is a device which responds to an electrical signal in such a manner that a magnetic pattern is created in a magnetisable medium.

b) Magnetic Tape:

→ Magnetic tape is composed of coating fine magnetic iron oxide particle $[Fe_2O_3]$ on a plastic ribbon. A typical type of size of tape is 12.7mm and 25.4 μm thick.

c) Reproducing Head:

→ Reproducing head is used to detect the stored magnetic pattern and to convert it back to digital electrical signal.

d) Tape Transport ~~devices~~ mechanism:

→ Tape transport mechanisms moves the magnetic tape along the recording head or reproducing head with a constant speed. The tape mechanism must be capable of handling the tape during various modes of operation without straining and wearing the tape.

e) Conditioning Device:

→ Conditioning device consists of amplifier and filters to modify signal to a format that can be properly recorded on a tape.

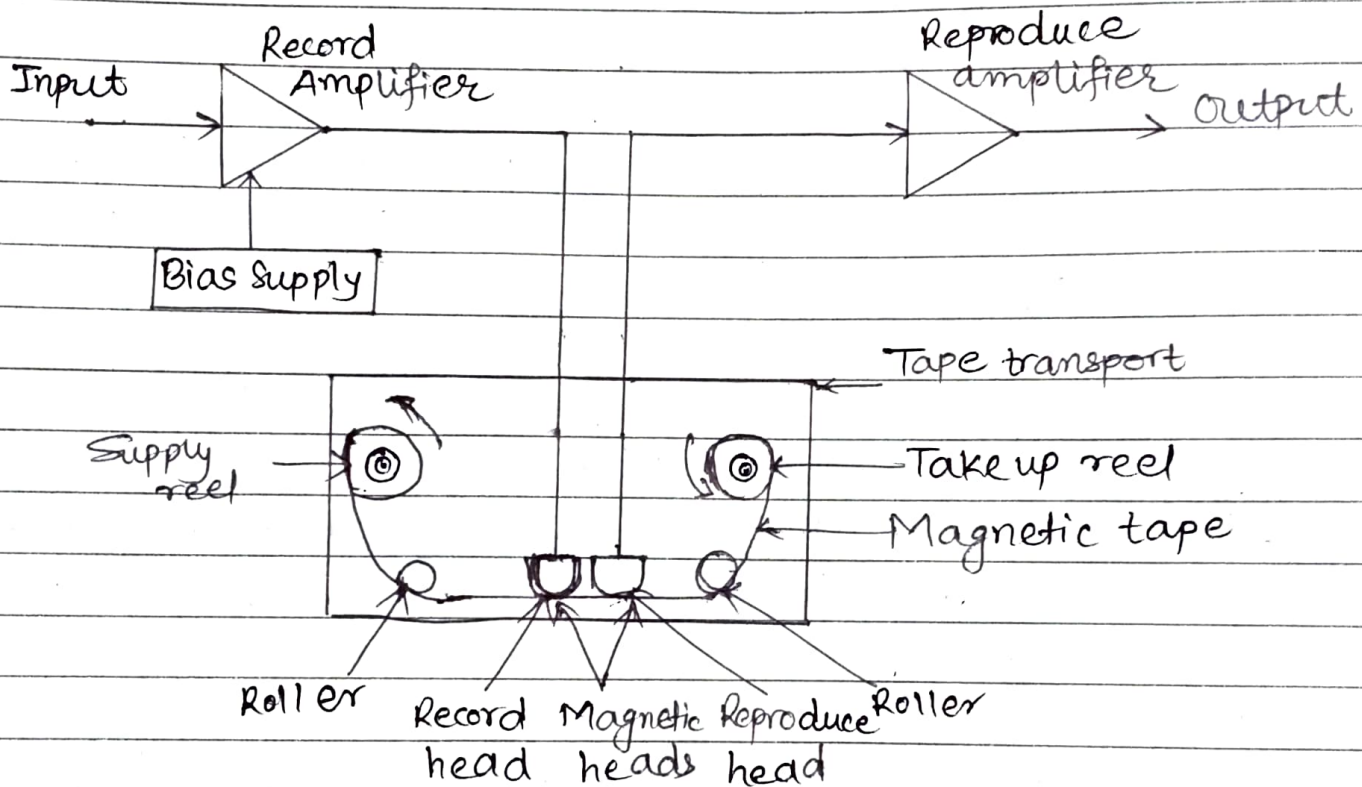


Fig. Magnetic Tape recorder

Operation:

- When a magnetic tape is passed through a recording head, any signal recorded on the tape appears as magnetic pattern dispersed in space along the tape.
- When the same tape is passed through a reproducing head, there will be induction of voltage in the tape by magnetization.

→ The induced voltage is proportional to the rate of change of flux linkages,

$$\text{i.e. } E \propto N \frac{d\phi}{dt}$$

where, N = no. of turns of winding.

→ Thus the output signal from the reproduce head is a derivative of the input signal.

→ The magnitude of the output signal is not only proportional to the flux recorded on the tape but also the frequency of the recorded signal.

⊗ Applications:

i> Communication surveillance and spying

ii> Medical research

iii> Data recording and analysis on missiles, air crafts and satellites.

iv> Industrial research and production monitoring and control.

* Advantages of magnetic tape recorder:

i > Wide frequency range from DC to several mHz.

ii > Low distortion

iii > It permits multichannel recording

iv > Exceeding high density of data points giving simplified storage and handling.

chapter-5

Data Transmission

1. Differences between serial Communication and Parallel Communication;

Ans

SN	Serial Communication	Parallel Communication
i>	In serial communication, a single link is used to transfer data from one point to another.	In parallel communication, multiple links are used to transfer data from one point to another.
ii>	Serial Communication is cost-efficient.	Parallel Communication is not cost-efficient.
iii>	In serial Communication, one bit is transferred at one clock pulse.	In parallel communication, 8-bits are transferred at one clock pulse.
iv>	Serial Communication is slow.	Parallel Communication is fast.
v>	Serial Communication is used for long distance.	Parallel Communication is used for short distance.
vi>	In serial communication, data flows in bi direction.	In parallel communication, data flows in multiple lines.
vii>	Serial Communication is full duplex.	Parallel Communication is half duplex.
viii>	In serial communication, the circuit used is simple.	In parallel communication, the circuit used is relatively complex.

2. Differences between synchronous and asynchronous data transmission.

Ans:

S.N.	Synchronous data transmission	Asynchronous data transmission
i>	In synchronous transmission, data is transferred in the form of blocks or frames.	In asynchronous transmission, data is transferred in the form of bytes or characters.
ii>	Synchronous transmission is fast.	Asynchronous transmission is slow.
iii>	Synchronous transmission is costly.	Asynchronous transmission is economical.
iv>	In synchronous transmission, the time interval of transmission is constant.	In asynchronous transmission, the time interval of transmission is random.
v>	Synchronous transmission needs precisely synchronized clocks.	Asynchronous transmission does not need synchronized clocks.
vi>	In synchronous transmission, there is no gap present between data.	In asynchronous transmission, there is gap present between data.
vii>	In synchronous transmission, errors are detected and corrected in real time.	In asynchronous transmission, errors are detected and corrected when the data is received.

3. Difference between half duplex and full duplex communication

Ans:

S.N.	Half duplex Communication	Full duplex Communication
i>	In half duplex communication, data can be sent or received at a time.	In full duplex communication, data can be sent and received simultaneously.
ii>	Direction of full half duplex communication is two directional but one at a time.	Direction of full duplex communication is two directional or and simultaneously.
iii>	Example: walkie-talkie etc.	Example: Telephone, mobile etc.
iv>	Block diagram: <pre> graph LR S[sender] <--> or R[Receiver] </pre>	Block diagram: <pre> graph LR S[sender] --> R[Receiver] R --> S </pre>

Q. Compare the wired channels used in data transmission:

eg:

S.N.	Twisted pair cable	Coaxial cable	Optical fiber cable
i>	In twisted pair cable, transmission of signals take place in the electrical form over metallic conducting wire.	In coaxial cable, transmission of signals take place in the electrical form over the inner conductor of cable.	In optical fiber cable, transmission of signals take place in the the optical form over a glass fiber.
ii>	Cheapest medium.	Moderate medium	Expensive medium.
iii>	Low bandwidth.	Moderately high bandwidth.	Very high bandwidth.
iv>	Attenuation is very high.	Attenuation is low.	Attenuation is very low.
v>	Installation is easy.	Installation is fairly easy.	Installation is difficult.
vi>	Noise immunity is low.	Higher noise immunity	Highest noise immunity.
vii>	It is affected by external magnetic field.	It is less affected by external magnetic field.	It is not affected by external magnetic field.

5. What are the ^{methods} types of data transmission? Explain the typical telemetering system with a block diagram.

Ans:

The methods of data transmission are:

- i) Hydraulic data transmission
- ii) Pneumatic data transmission
- iii) Electrical or electronic data transmission.

→ Telemetry is the process of collecting and transmitting data from remote or inaccessible sources to a monitoring or control system.

→ It is commonly used in various fields such as aerospace, healthcare and industrial setting.

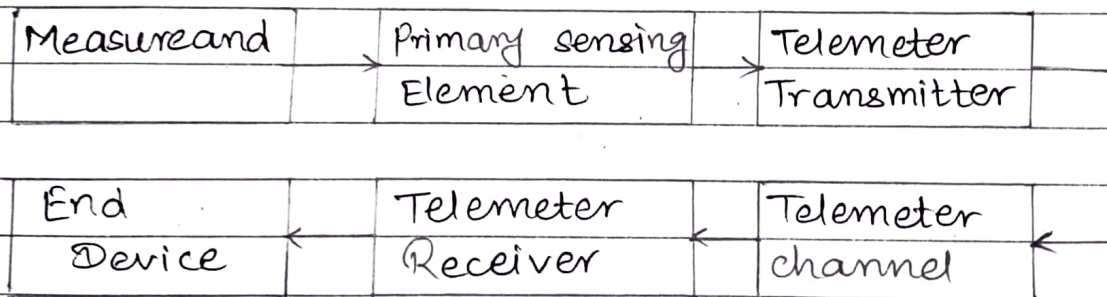


Fig. Block diagram of telemetering system

→ The primary detector and end device of the telemetering system have the same functional positions as in any general measurement system.

- The function of the telemeter transmitter is to convert the output of a primary detector into a related quantity which can be transmitted over the telemeter channel.
- The function of telemeter receiver at the remote location is to convert the transmitted signal into a related suitable quantity.

* Application of telemetry system:

- i> Meteorology
- ii> Motor racing
- iii> Transportation
- iv> Agriculture
- v> Defense, Space and resource exploration
- vi> Space Science
- vii> Flight testing

chapter-1 Introduction to Instrumentation System

1) Explain instrumentation system and also the generalized components of instrumentation system with the help of block diagram.

Ans → An instrumentation system is a way to carry out the measurement.

→ An instrumentation system is a collection of instruments used to measure, monitor and control a process.

→ The Generalized block diagram of an instrumentation system is shown below:

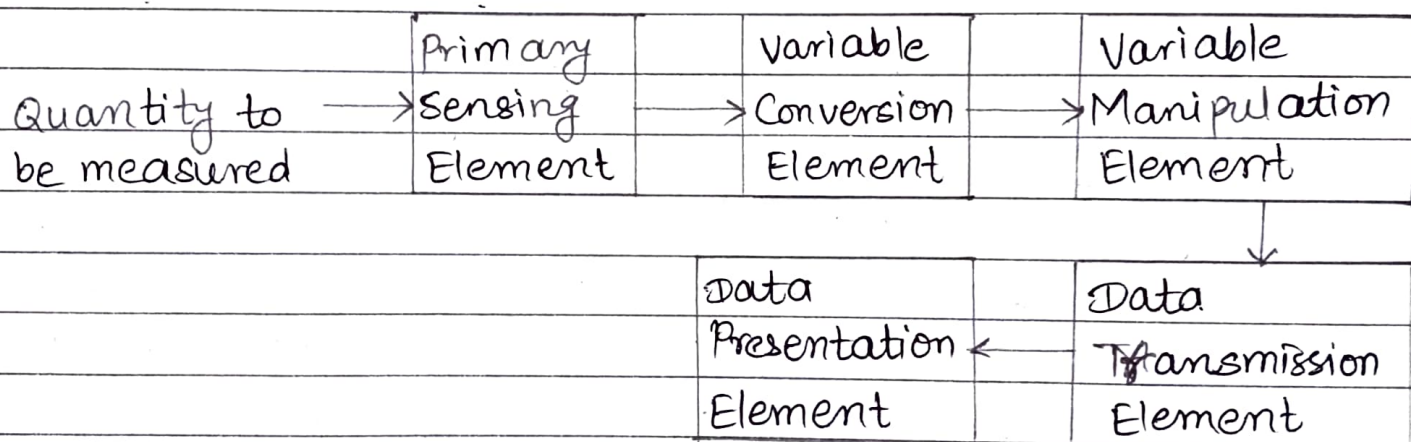


Fig. Generalized block diagram of an instrumentation system.

i) Primary Sensing Element:

→ The quantity under measurement makes its first contact with primary sensing element known as transducer.

→ The transducer is a device which converts physical parameters into electrical signals.

→ For example: Microphone, LDR etc.

ii) Variable Conversion Element:

- The output of primary sensing element may be electrical signal of any form.
- This input is not suited to the system. so variable conversion element is required to convert these signals.

iii) Variable Manipulation Element:

- The output variable conversion element is required to manipulate.
- The function of variable manipulation element is to manipulate the signal present to it preserving the original nature of the signal.

iv) Data Transmission Element:

- The output of variable manipulation element is needed to transmit from one point to another.
- The element that performs this task is called data transmission element.

v) Data Presentation Element:

- After data transmission element, the information about the quantity being measured should be displayed approximately for monitoring, control and analysis purpose.

2. Define signals. Explain the different types of signals used in instrumentation system.

Ans A signal is defined as a function of one or more independent variables that contains information about the nature of some phenomenon.

Types of signals used in instrumentation system:

i) Continuous-time and discrete-time signals:

→ Continuous-time signal is defined for all time t .

→ Discrete-time signal is defined only at discrete time.

ii) Even and odd signals:

→ Even signal is identical to its time-reversed counterpart.

→ Odd signal is ~~identical~~ not identical to its time-reversed counterpart.

iii) Periodic and non-periodic signal.

iv) Deterministic and Random signals.

v) Real and complex signals.

vi) Energy and Power signal.

vii) Analog and Digital signal.

viii > Unit step signal

ix > Ramp signal

x > Unit impulse signal

xi > causal, anti-causal and non-causal signal.